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Wireshark capture lab

During this lab, I created a self-contained virtual machine environment for network analysis and installed Wireshark inside the VM. One challenge I encountered was making sure virtualization was enabled on the host computer and that the VM had enough resources assigned to run smoothly. Because of connectivity issues and a previous encounter with malware, I decided to use Kali-Linux, on Hyper-V. I also had to verify that the virtual network adapter was active so Wireshark could detect traffic from the VM. After updating Wireshark and allowing the packet capture driver to be installed, I tested the setup by visiting safe websites and capturing basic network activity.

Analyzing traffic inside a virtual machine is safer than using the host system because the VM creates separation between the lab environment and the main computer. If a capture file, website, or later lab activity exposes suspicious traffic or malware-like behavior, the risk is more contained. The VM can also be reset, deleted, or restored without damaging the host operating system. This makes the environment better for repeatable cybersecurity labs and controlled packet analysis.